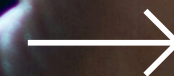


TIDY01: Fundamentals of Tidyverse I

Presented by Gabriel Rodrigues Palma



Tidyverse for Ecologists
(TIDY01)



Day 2 (9:30 – 17:30)

Data Cleaning
(9:35 – 11:05)

Grammar of Graphics
(13:30 – 15:00)

Data Filtering and
Selecting (11:05 – 12:30)

Tidyverse for Ecologists
(TIDY01)



Grammar of Graphics

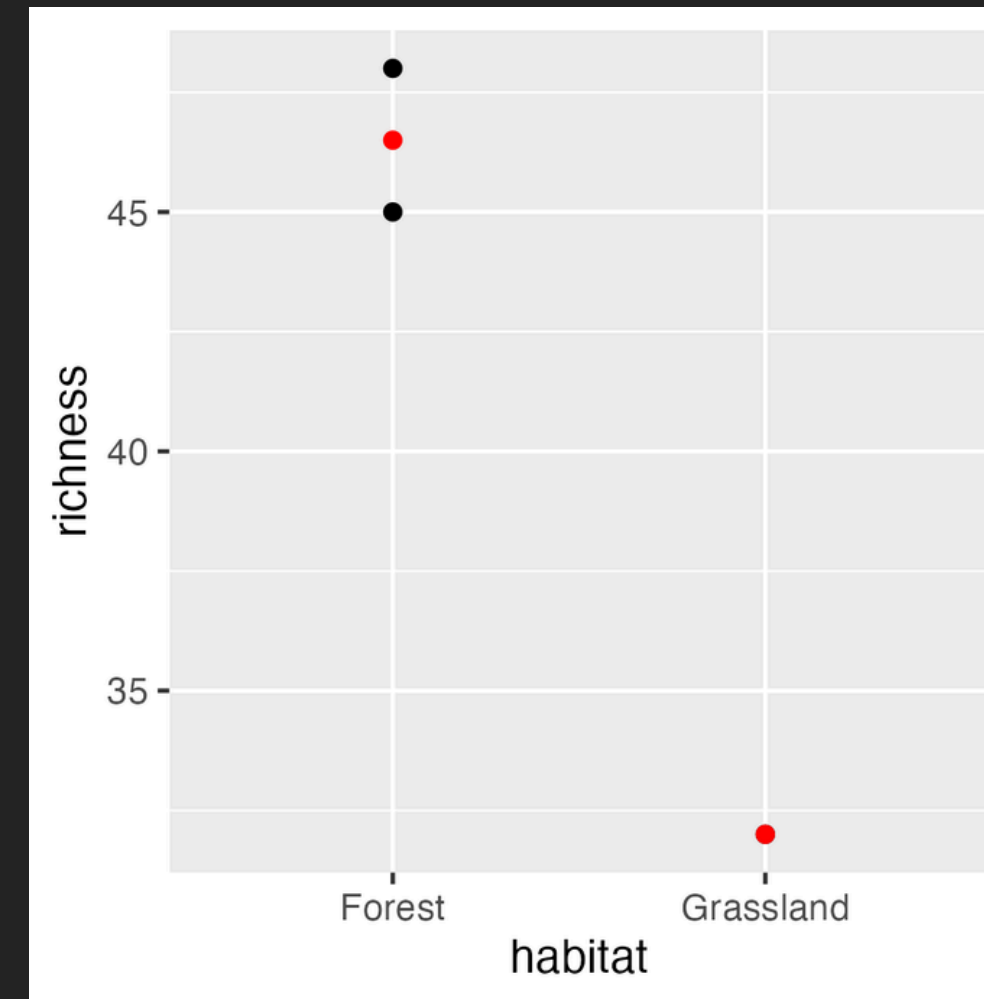
Tidyverse for Ecologists
(TIDY01)



Grammar of Graphics

R Input

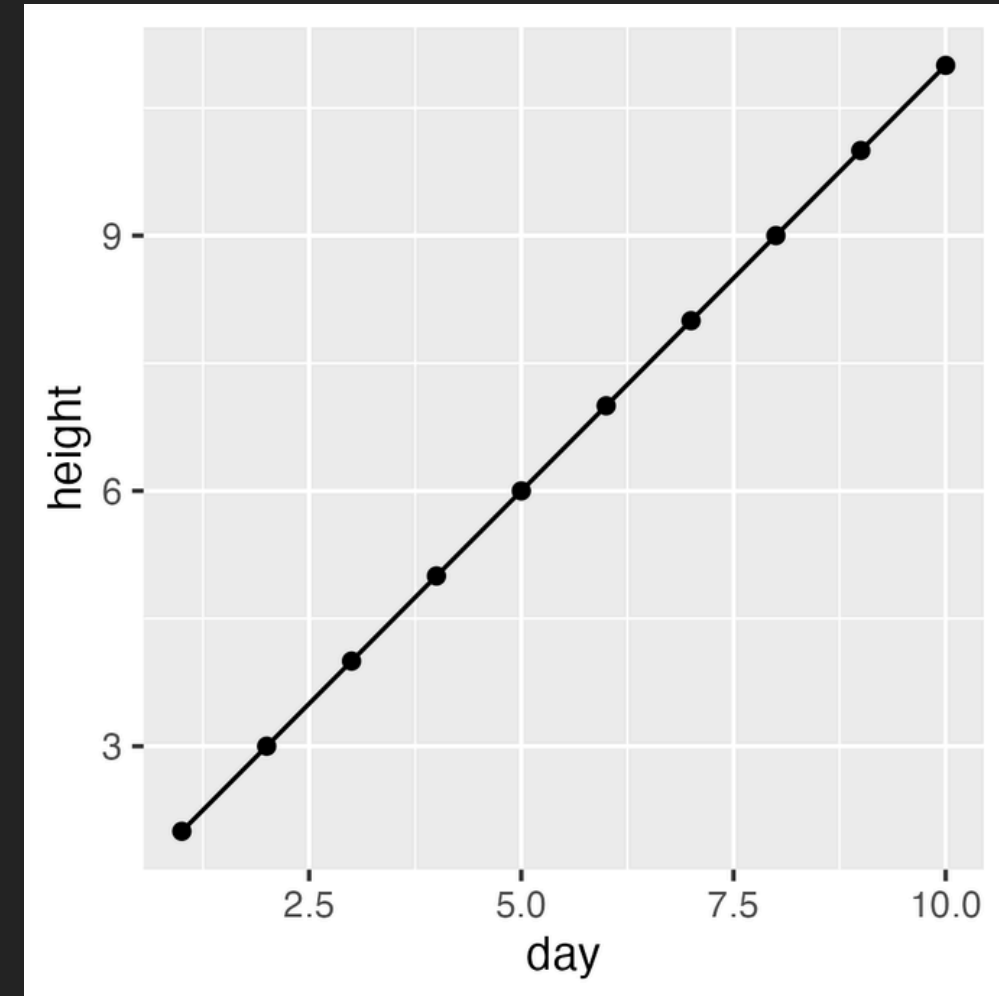
```
# Statistical summary example
biodiversity <- data.frame(
  habitat = c('Forest', 'Grassland', 'Forest'),
  richness = c(45, 32, 48)
)
ggplot(biodiversity, aes(x = habitat, y = richness)) +
  geom_point() + stat_summary(fun = mean, geom = 'point', color = 'red')
```



Grammar of Graphics

R Input

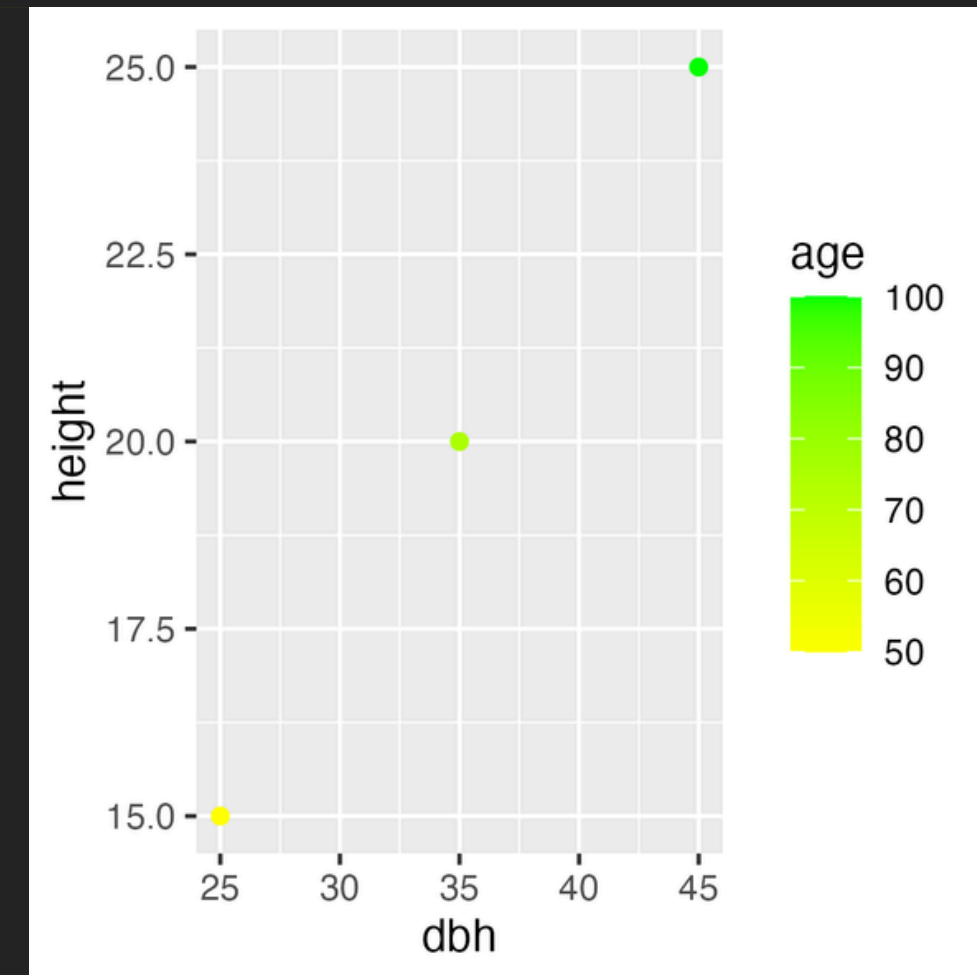
```
# Multiple layers example
plant_growth <- data.frame(
  day = 1:10,
  height = c(2, 3, 4, 5, 6, 7, 8, 9, 10, 11)
)
ggplot(plant_growth, aes(x = day, y = height)) +
  geom_point() + geom_line()
```



Grammar of Graphics

R Input

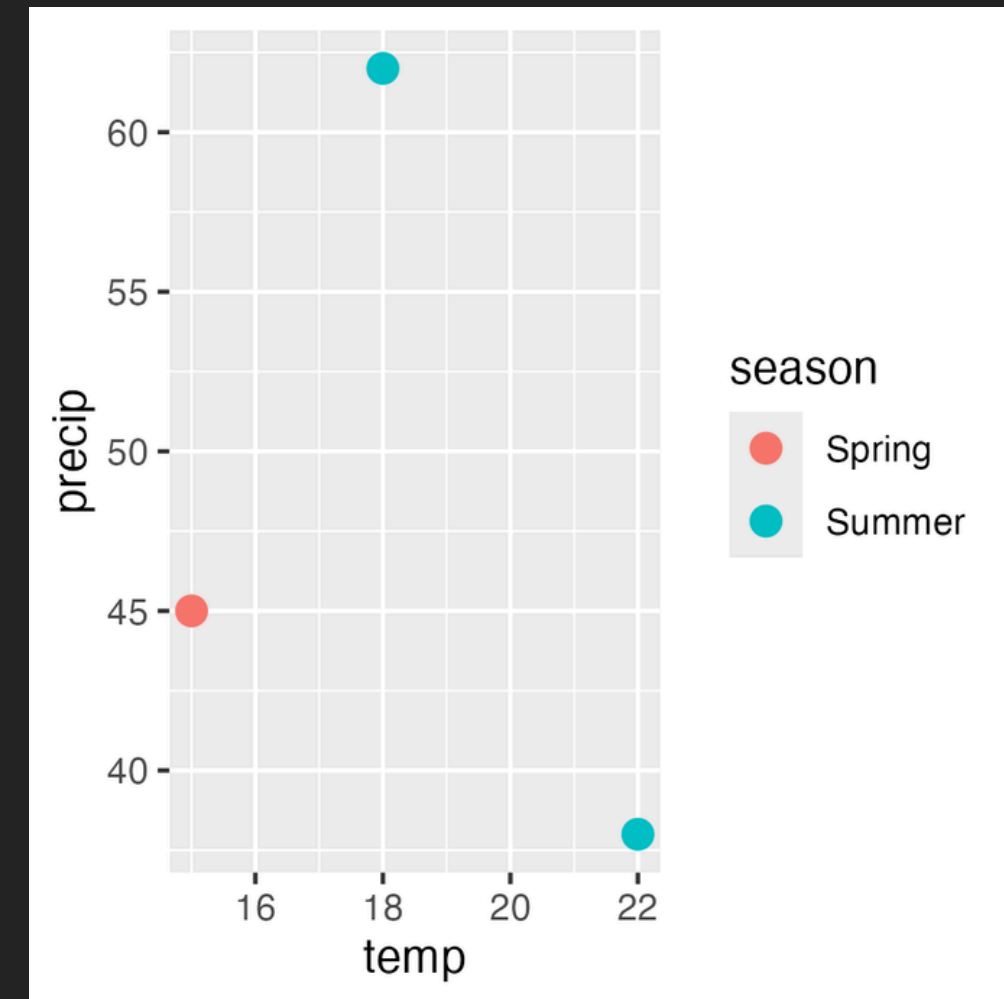
```
# Custom scales example
tree_data <- data.frame(
  dbh = c(25, 35, 45),
  height = c(15, 20, 25),
  age = c(50, 75, 100)
)
ggplot(tree_data, aes(x = dbh, y = height, color = age)) +
  geom_point() + scale_color_gradient(low = 'yellow', high = 'green')
```



Grammar of Graphics

R Input

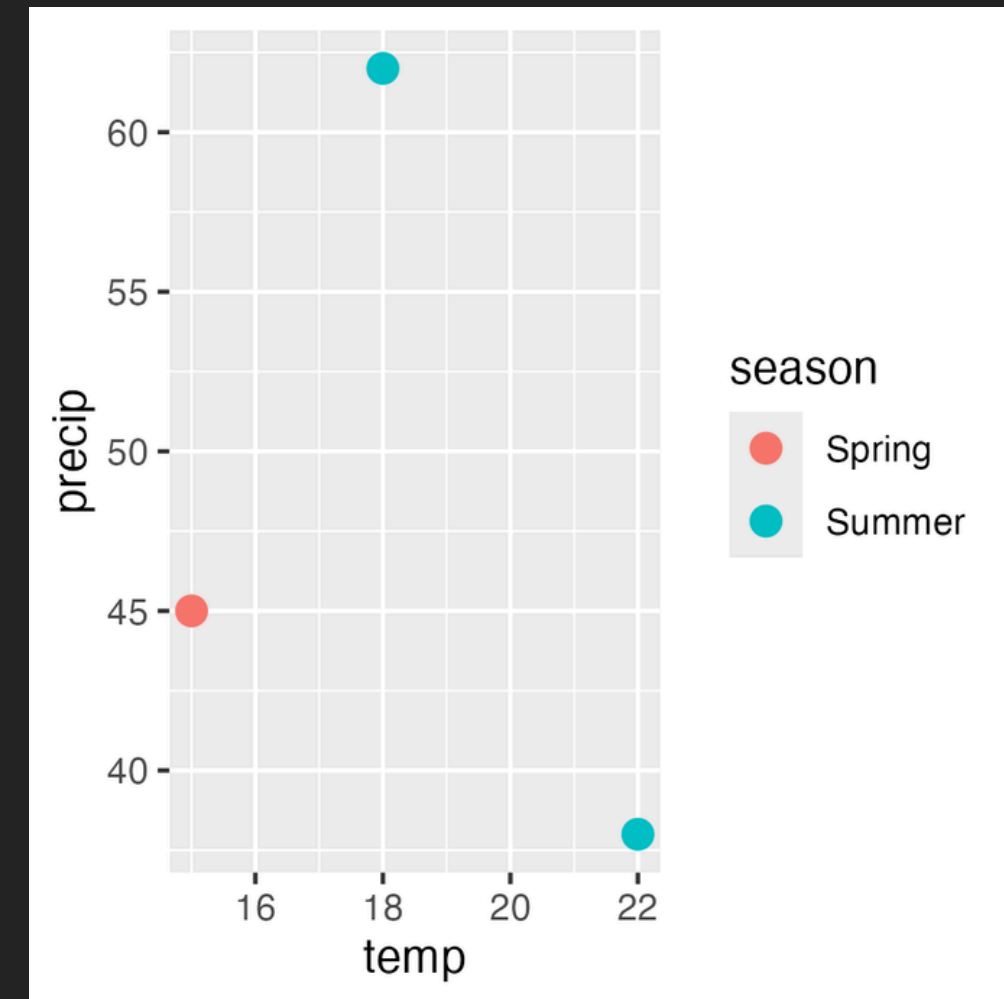
```
# Adding color aesthetics
climate <- data.frame(
  temp = c(15, 18, 22),
  precip = c(45, 62, 38),
  season = c('Spring', 'Summer', 'Summer')
)
ggplot(climate, aes(x = temp, y = precip, color = season)) +
  geom_point(size = 3)
```



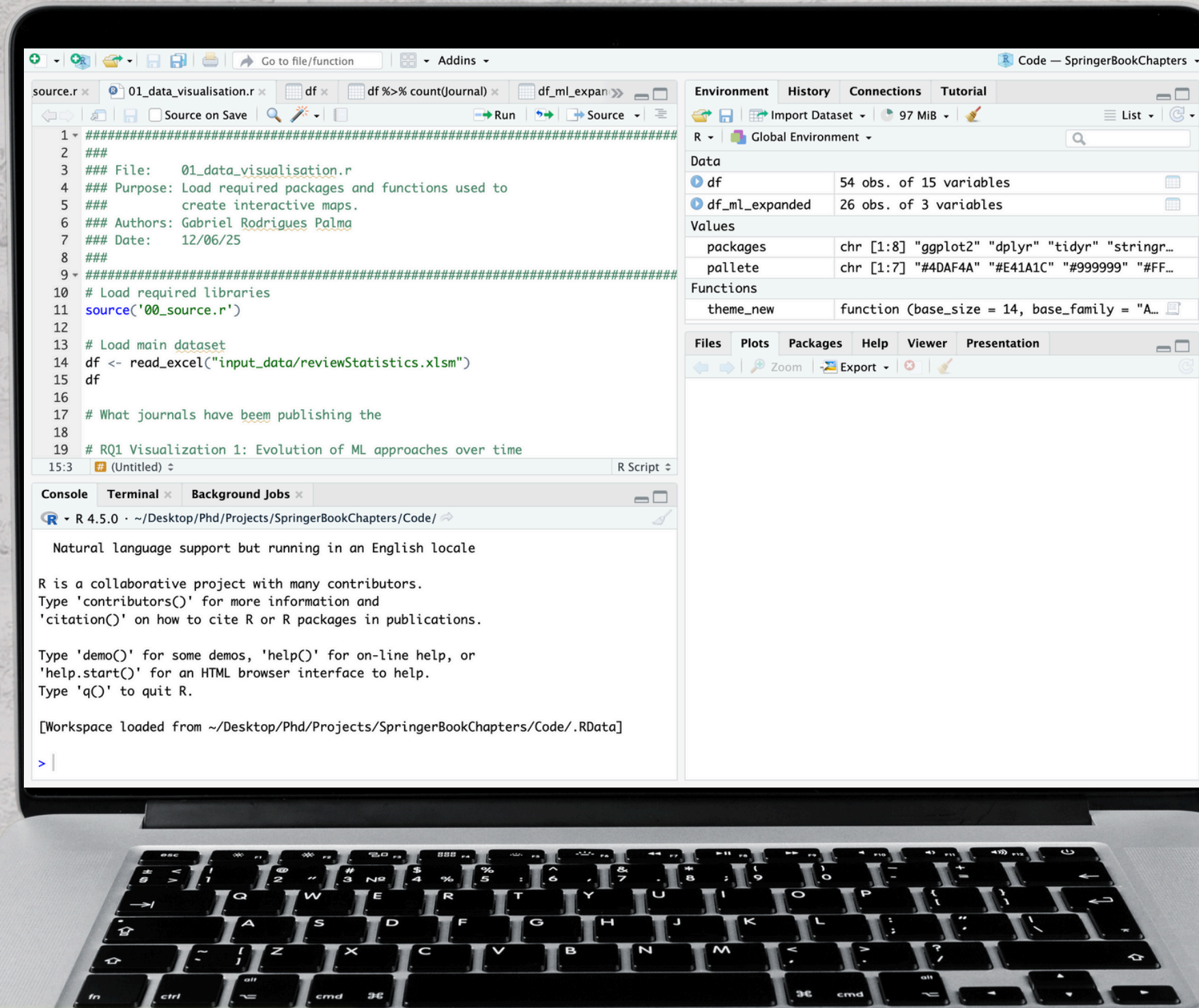
Grammar of Graphics

R Input

```
# Basic ggplot structure
library(ggplot2)
species_data <- data.frame(
  species = c('Oak', 'Pine'),
  count = c(25, 18)
)
ggplot(species_data, aes(x = species, y = count)) +
  geom_point()
```



Grammar of Graphics



Tidyverse for Ecologists
(TIDY01)



Day 2 (9:30 – 17:30)

Data Cleaning
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Data Filtering and
Selecting (11:05 – 12:30)

Basic Plots
(15:00 – 16:30)

Tidyverse for Ecologists
(TIDY01)



Total Confirmed
566,269

Basic Plots

Confirmed Cases by
Country/Region/Sovereignty

92,932 US

99,997 China

NORTH
AMERICA

Machine Learning
using Python (TIDY01)

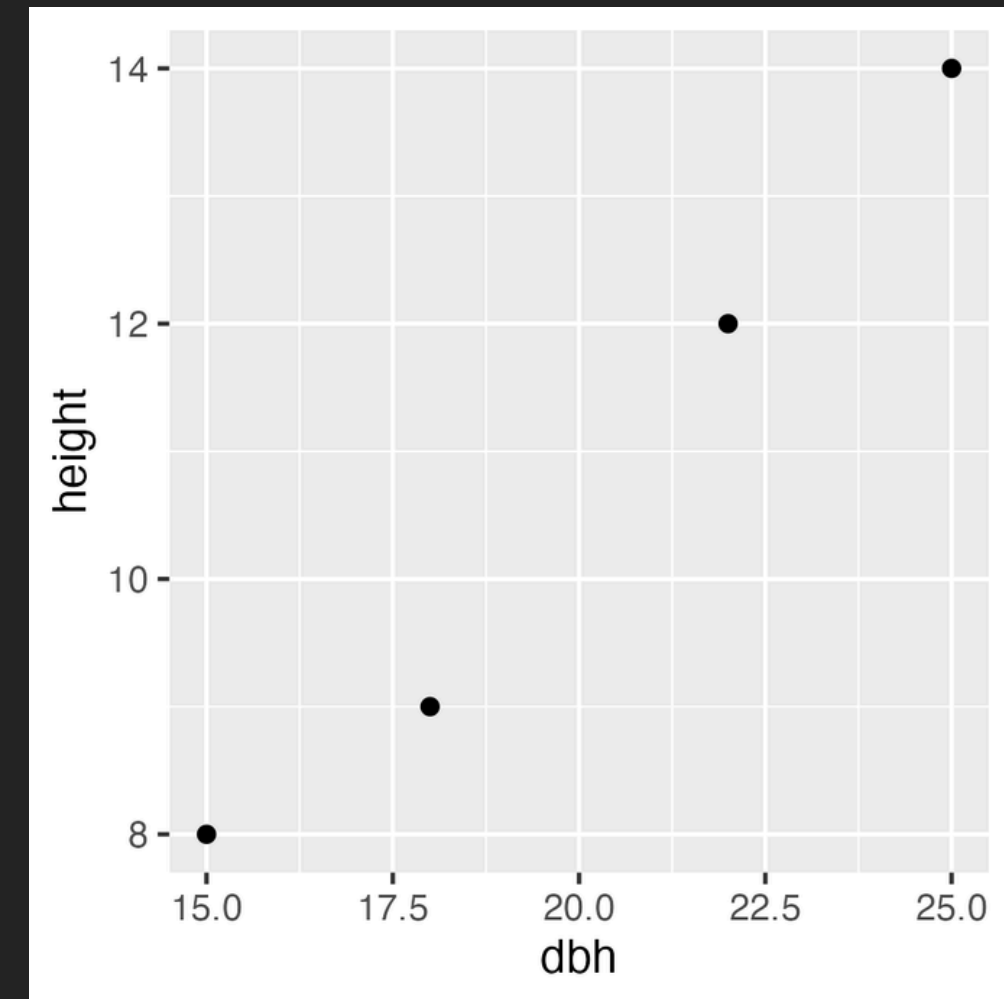
R stats



Basic Plots

R Input

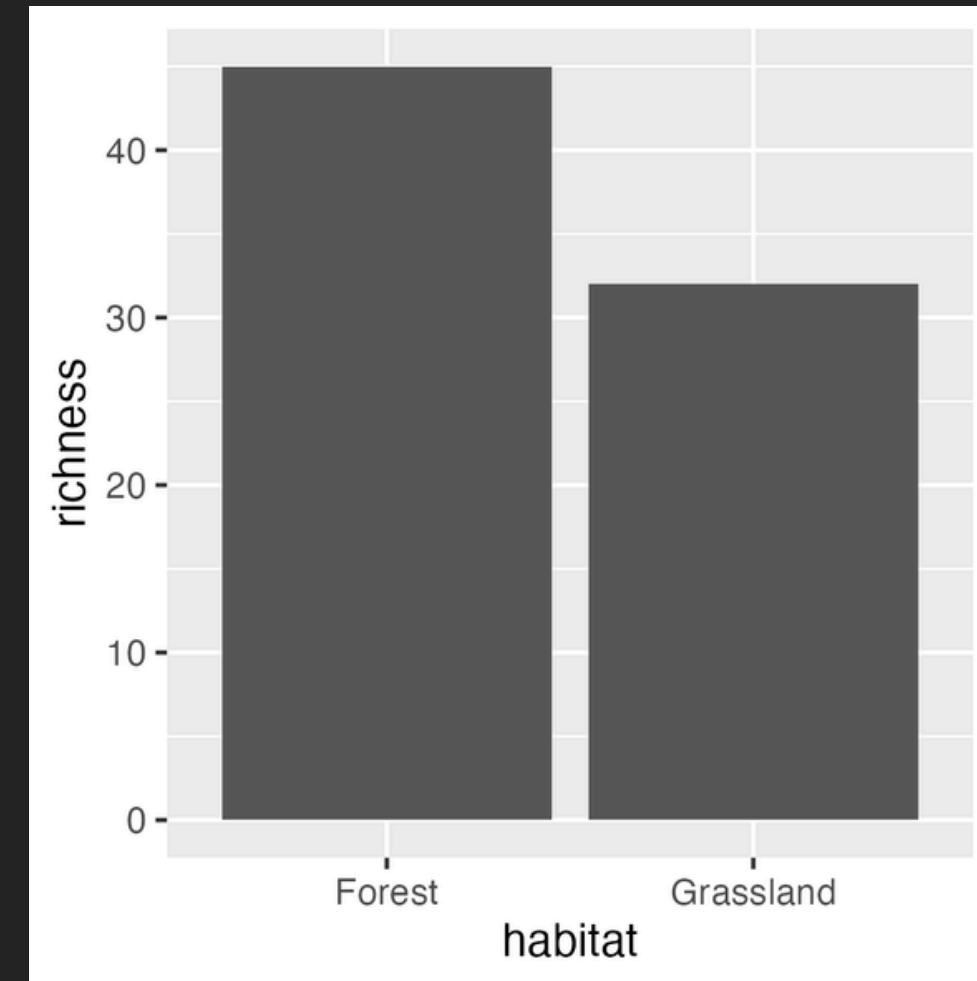
```
# Basic scatterplot
tree_data <- data.frame(
  dbh = c(15, 22, 18, 25),
  height = c(8, 12, 9, 14)
)
ggplot(tree_data, aes(x = dbh, y = height)) +
  geom_point()
```



Basic Plots

R Input

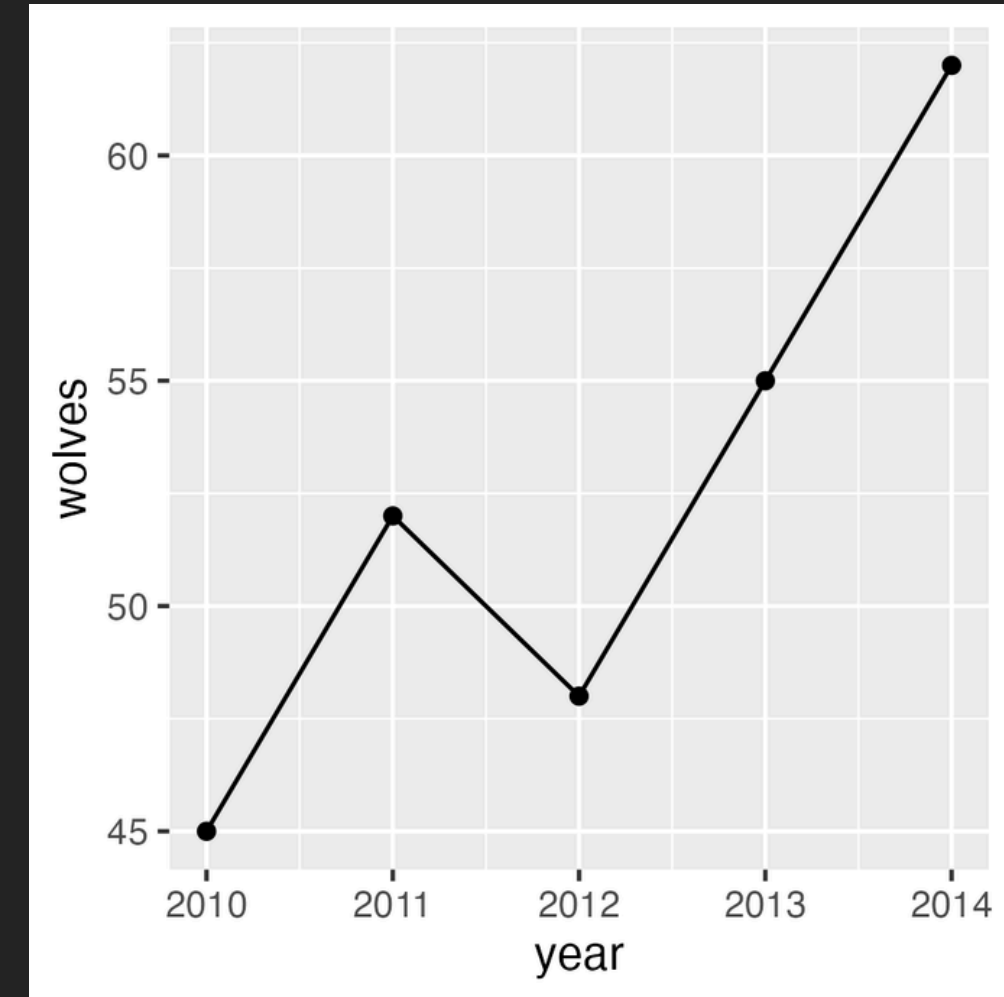
```
# Bar chart with species counts
species_counts <- data.frame(
  habitat = c('Forest', 'Grassland'),
  richness = c(45, 32)
)
ggplot(species_counts, aes(x = habitat, y = richness)) +
  geom_col()
```



Basic Plots

R Input

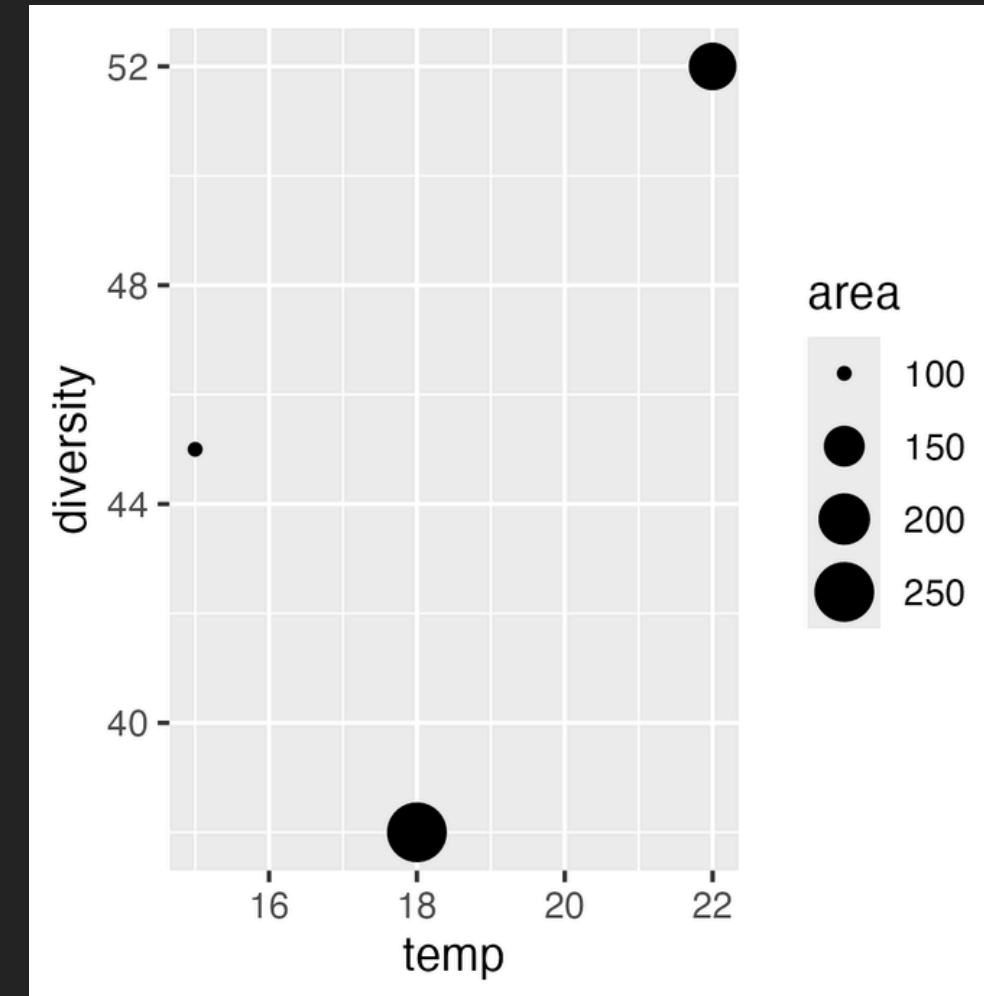
```
# Line graph for population trends
population <- data.frame(
  year = 2010:2014,
  wolves = c(45, 52, 48, 55, 62)
)
ggplot(population, aes(x = year, y = wolves)) +
  geom_line() + geom_point()
```



Basic Plots

R Input

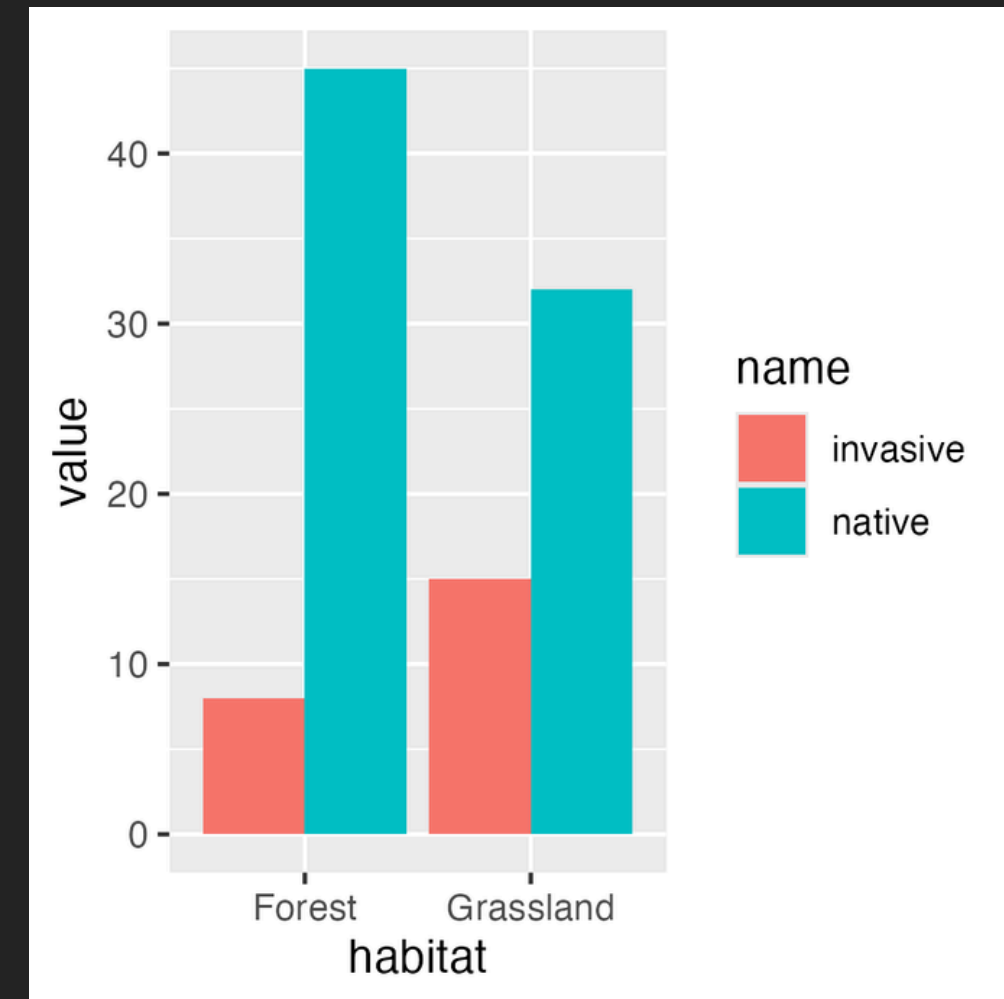
```
# Bubble plot with multiple variables
ecosystem <- data.frame(
  temp = c(15, 18, 22),
  diversity = c(45, 38, 52),
  area = c(100, 250, 180)
)
ggplot(ecosystem, aes(x = temp, y = diversity, size = area)) +
  geom_point()
```



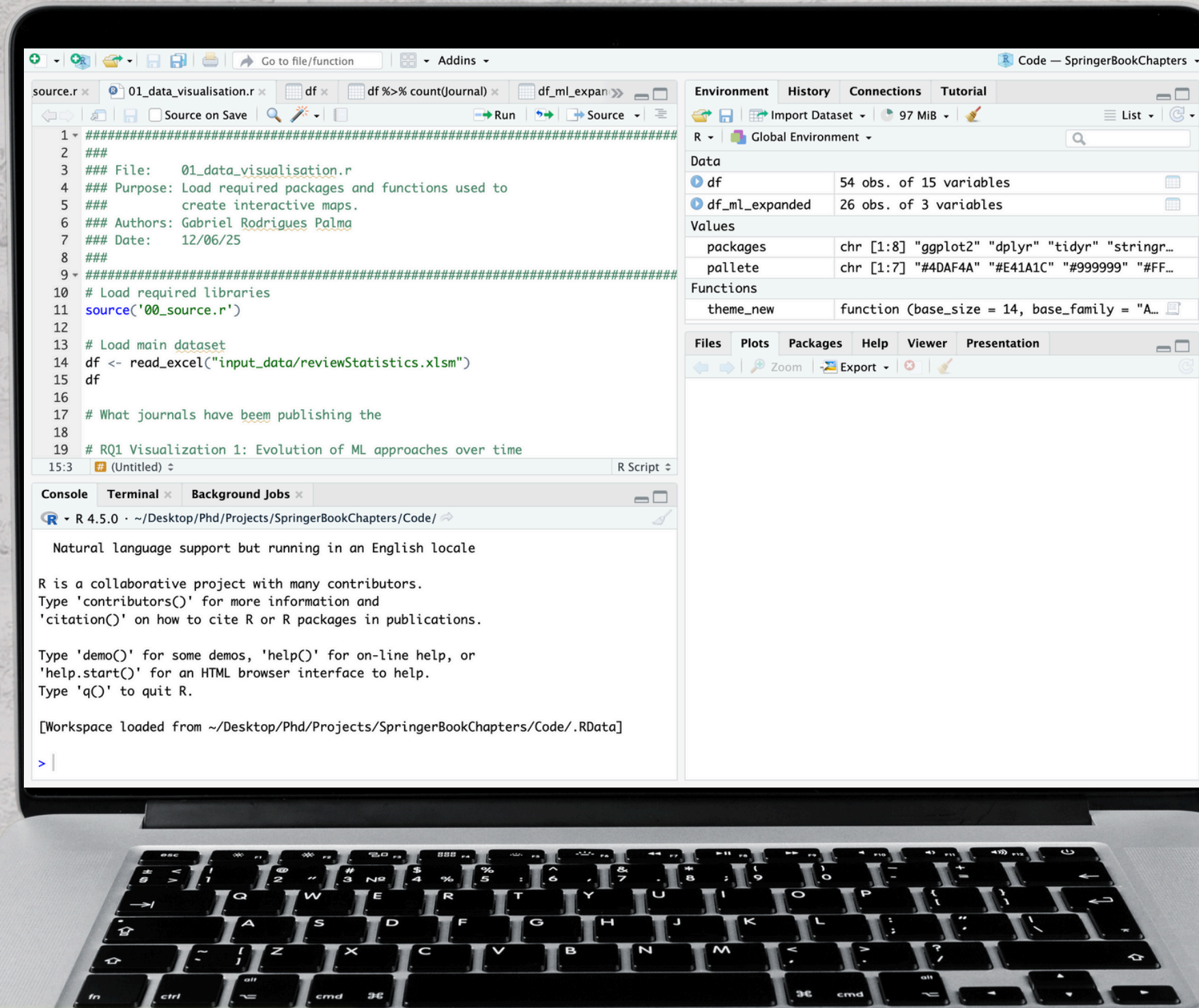
Basic Plots

R Input

```
# Grouped bar chart
biodiversity <- data.frame(
  habitat = c('Forest', 'Grassland'),
  native = c(45, 32),
  invasive = c(8, 15)
)
tidyr::pivot_longer(biodiversity, -habitat) %>%
  ggplot(aes(x = habitat, y = value, fill = name)) +
  geom_col(position = 'dodge')
```



Basic Plots



Tidyverse for Ecologists
(TIDY01)



Day 2 (9:30 – 17:30)

Data Cleaning
(9:35 – 11:05)

Grammar of Graphics
(13:30 – 15:00)

Layers, Scales, and
Faceting (16:30 – 17:30)

Data Filtering and
Selecting (11:05 – 12:30)

Basic Plots
(15:00 – 16:30)

Tidyverse for Ecologists
(TIDY01)



Layers, Scales, and Faceting

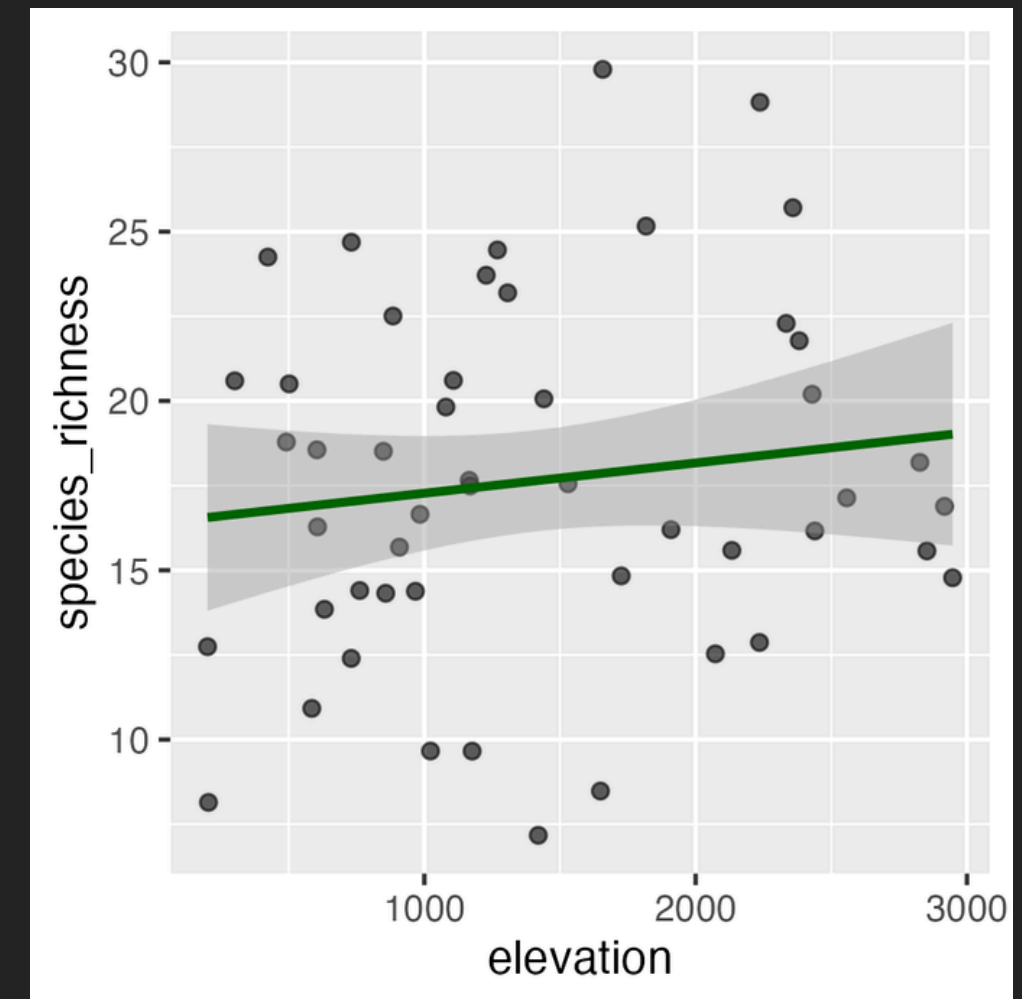
Machine Learning
using Python (TIDY01)



Layers, Scales, and Faceting

R Input

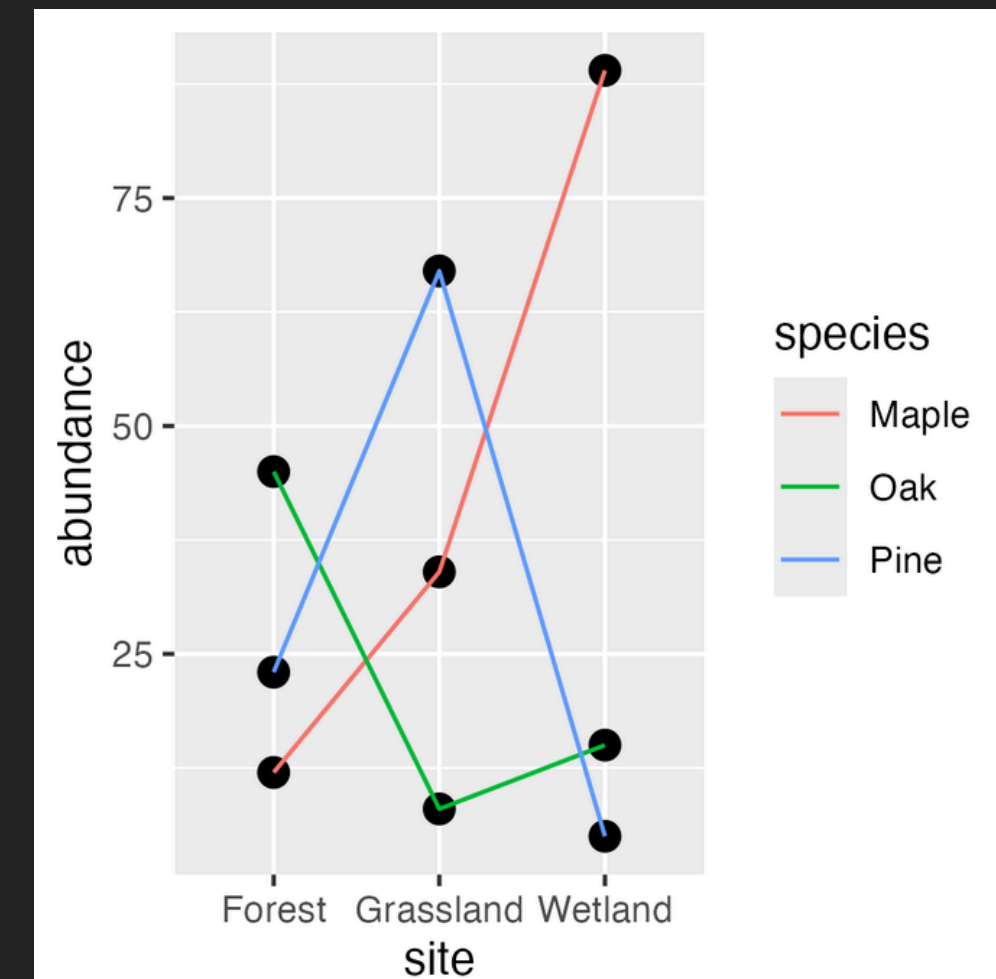
```
# Statistical layers for biodiversity patterns
biodiversity_data <- data.frame(
  elevation = runif(50, 100, 3000),
  species_richness = 25 - 0.005 * runif(50, 100, 3000) + rnorm(50, 0, 3)
)
ggplot(biodiversity_data, aes(x = elevation, y = species_richness)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, color = "darkgreen")
```



Layers, Scales, and Faceting

R Input

```
# Basic layer structure with species data
library(ggplot2)
species_data <- data.frame(
  site = rep(c("Forest", "Grassland", "Wetland"), each = 3),
  species = rep(c("Oak", "Pine", "Maple"), 3),
  abundance = c(45, 23, 12, 8, 67, 34, 15, 5, 89)
)
ggplot(species_data, aes(x = site, y = abundance)) +
  geom_point(size = 3) +
  geom_line(aes(group = species, color = species))
```



Layers, Scales, and Faceting

source.r x 01_data_visualisation.r x df x df %>% count(Journal) x df_ml_expan x

Go to file/function Addins

Source on Save Run Source

```
1 #####
2 ###
3 ### File: 01_data_visualisation.r
4 ### Purpose: Load required packages and functions used to
5 ### create interactive maps.
6 ### Authors: Gabriel Rodrigues Palma
7 ### Date: 12/06/25
8 ###
9 #####
10 # Load required libraries
11 source('00_source.r')
12
13 # Load main dataset
14 df <- read_excel("input_data/reviewStatistics.xlsm")
15 df
16
17 # What journals have been publishing the
18
19 # RQ1 Visualization 1: Evolution of ML approaches over time
20
21 15:3 (Untitled) R Script
```

Environment History Connections Tutorial

Import Dataset 97 MiB List

R Global Environment

Data

df	54 obs. of 15 variables
df_ml_expanded	26 obs. of 3 variables

Values

packages	chr [1:8] "ggplot2" "dplyr" "tidyr" "stringr..."
pallette	chr [1:7] "#4DAF4A" "#E41A1C" "#999999" "#FF..."

Functions

theme_new	function (base_size = 14, base_family = "A..."
-----------	--

Files Plots Packages Help Viewer Presentation

Zoom Export

Console Terminal Background Jobs

R R 4.5.0 ~ / Desktop / Phd / Projects / SpringerBookChapters / Code /

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~ / Desktop / Phd / Projects / SpringerBookChapters / Code / .RData]

> |